The Topology of Reachable Nature: Persistent Homology and the Resilience of Green-Space Access in Cities of Contrasting Morphology

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Abstract

Sustainable Development Goal 11.7 calls for universal access to green and public space, yet accessibility is almost always quantified as a static snapshot that says nothing about how access degrades when the street network is disrupted. We introduce a *topological resilience metric* for accessibility: the Wasserstein distance between the persistent-homology diagram of a city's green-space accessibility field before and after network disruption, traced as a function of disruption intensity ρ. The area under this degradation curve and its critical transition ρ^* summarise how robustly a place keeps its *locations* close to nature. This is one facet of SDG 11.7, the pedestrian network's reach to green space, and not its population-coverage dimension. Applying the metric to five morphologically contrasting cities (İstanbul, Barcelona, Amsterdam, Bogotá, Phoenix) at the level of 2,603 intra-urban districts, we find that street-network morphology is consistently associated with the resilience of the *connected* green-space access structure. Two descriptors are robust: in a district-level regression with city fixed effects, more circuitous routing and longer street segments are associated with greater resilience, and both survive spatial-error and spatial-lag models that absorb the residual spatial autocorrelation. A third, higher orientation entropy, is associated with less resilience. All three signs replicate in an eight-city extension (3,999 districts, adding Singapore, Nairobi and Vienna), where, with eight clusters rather than five, all three— including orientation entropy—survive city-clustered standard errors; the signs are also stable across random and targeted disruption and district resolutions spanning a 28$\times$ range. The metric itself passes a within-city criterion check: it correlates positively with a distance-based accessibility-degradation benchmark (Spearman 0.36) and, using the open GHS-POP grid, with a population-weighted one (0.35), while adding information beyond both, with the scope that it captures the resilience of the *connected* served structure and not total disconnection. The signal lives in dimension-0 homology, the merging of well-served regions, and not in enclosed "access deserts," a finding that emerges on real street networks.

1 Introduction

Target 11.7 of Sustainable Development Goal 11 commits states to provide "universal access to safe, inclusive and accessible, green and public spaces" [18]. Planners typically measure progress with catchment or proximity analyses: the share of residents within a walking threshold of a park. Such measures are static. They describe access on an ordinary day and are silent about *resilience*—how access to nature holds up when the pedestrian network is degraded by a flood, an earthquake, or the closure of a critical bridge or street. Two gaps follow. First, a city can score well on average proximity yet collapse under modest disruption; average accessibility does not reveal this fragility. Second, accessibility work rarely connects the *shape* of access to urban *form*, which is precisely the architecture–mathematics question that motivates this collection.

Persistent homology (PH) offers a principled, multi-scale, coordinate-free language for the "holes" and connected pieces of a coverage pattern [9, 13]. PH has been used to detect coverage gaps for amenities such as polling sites [11] and healthcare [10], to read the topology of urban congestion [19], and to quantify displacement under urban expansion [14]. What is *not* novel is using PH to find accessibility "holes" for an amenity. Our contribution is to move from *detecting* coverage gaps to *measuring the resilience of the access topology* and to *linking that resilience to urban form.* A scope note up front: we measure *network* accessibility, the pedestrian graph's spatial reach to green space with every network node weighted equally, and not *population* accessibility, which would weight locations by the residents they serve. The two coincide only where population is uniform. Our claims concern the network's structural capacity to keep places near nature, not how many people retain access.

Contributions.

- **A topological resilience metric** (the mathematics). We define $D(\rho)$, the Wasserstein distance between the persistent-homology diagram of the green-space accessibility field before and after disruption of intensity ρ, and summarise it by an area-under-curve (AUC) and a critical transition ρ^* (Section 4).
- **First topological treatment of SDG 11.7 green-space access** in an architecture–mathematics framing.
- **A district-level morphology↔resilience result** across 2,603 districts in five cities, with city fixed effects, showing a consistent association between street-network morphology and access resilience whose sign is stable across disruption models and district resolutions (Section 6).

2 Related work

TDA of spatial systems. Feng et al. [9] survey persistent homology for spatial data; Otter et al. [13] give the computational foundations. Within urban analysis, Wu et al. [19] read traffic congestion as a barcode, and a line of "resource coverage" work applies sublevel or network-distance filtrations to amenity access: polling places [11] and sexual-health services [10]. Rodríguez Vázquez and Cuerno [14] use PH to quantify displacement under urban expansion. These studies establish that PH can *locate* coverage holes. We differ on the object of study: we do not report where the deserts are, but how the whole access topology *degrades* under disruption, and we regress that degradation on morphology.

Accessibility and equity. A large body of work measures green-space access as a static field, whether by simple footpath proximity [4] or by catchment methods such as the Gaussian two-step floating catchment area [16], often to diagnose inequities in who lives close to nature [1]. Our accessibility field (Section 4) is built the same way (network walking time to the nearest green space) but is then read topologically and stress-tested, rather than thresholded once.

Street-network morphology. We characterise form with standard descriptors computed from OpenStreetMap via OSMnx [2]: intersection density, circuity, orientation entropy, and mean street-segment length. Relating these to a resilience outcome is the architecture–mathematics bridge of this paper.

3 Mathematical background

Let $G = (V, E)$ be the pedestrian network and $f : V \to \mathbb{R}$ a scalar field giving, at each node, the network walking time to the nearest green-space access point. The *sublevel-set filtration* of f builds a growing simplicial complex $K_t = \{\, \sigma : \max_{v \in \sigma} f(v) \le t \,\}$. As the threshold t (walk-minutes) rises, well-served regions appear and merge. Persistent homology records, for each topological feature, a birth b and death d; the multiset of points (b, d) is the *persistence diagram* $\mathrm{Dgm}_k(f)$ in homological dimension k [8]. Dimension 0 tracks connected components (the merging of well-served regions); dimension 1 tracks loops (enclosed voids, i.e. access "deserts").

The distance between two diagrams is measured by an optimal matching. The 1-Wasserstein distance is

$$W_1(X, Y) = \min_{\gamma} \sum_{x \in X} \|x - \gamma(x)\|,$$

minimised over bijections γ between the diagrams (with matching to the diagonal allowed). These diagram distances are stable to perturbations of f: the bottleneck distance by the classical stability theorem [5], and the 1-Wasserstein distance we use by the L_p-stability result of Cohen-Steiner et al. [6]. A small change in the field produces a small change in the distance. Stability is a mathematical well-posedness property, not a validation that diagram distance measures *urban* resilience; we use it only to justify that $D(\rho)$ responds continuously and monotonically to progressive network damage, and we treat the metric throughout as a *proposed proxy* for how the access topology holds up, to be validated against independent resilience evidence in future work rather than as ground truth. We compute PH with GUDHI [17] and diagram distances with `persim`.

4 Methodology

Data. For each city we take the pedestrian network and green spaces from OpenStreetMap via OSMnx [2], and we fix the analysis extent to the city's *urban centre* polygon from the GHS Urban Centre Database [12] so that extents are comparable across countries. Green spaces are the OSM classes `leisure=park|garden|recreation_ground`, `landuse=grass|recreation_ground`, `place=square` and `natural=wood`; their centroids, snapped to the nearest network node, are the access points. We call these *green* spaces, not *public* spaces: OSM tags do not certify public accessibility, so a private garden, a fenced lawn or an impenetrable wood can enter the set. This is a deliberately inclusive definition: it merges spaces of very different type, size and usability (a pocket garden, a city square and a large wood all count equally), and it takes the centroid as a single access point rather than modelling entrances or area. A sensitivity analysis under a strict, accessibility-conservative subset (dropping wood, grass and gardens; keeping parks, recreation grounds and squares) is reported below. We treat every green space as an equivalent point source; distinguishing quality, size, or public/private status is left to future work (see Limitations).

Accessibility field. We assign each node its network walking time to the nearest green-space access point by multi-source Dijkstra, giving the field f (Fig. 1). We convert length to time at a constant 4.8 km/h (1.33 m/s), which sits within the comfortable adult gait speed of 1.27–1.46 m/s reported by Bohannon [3]. The exact value does not matter for our conclusions: speed enters only through time = length/speed, so a different speed rescales the whole field, and hence every diagram, Wasserstein distance and AUC, by the same constant. The regression signs, significances and the city ranking are therefore invariant to the speed choice (if the persistence threshold is rescaled with speed the invariance is exact). We confirm this empirically: on Amsterdam, recomputing at 4.0

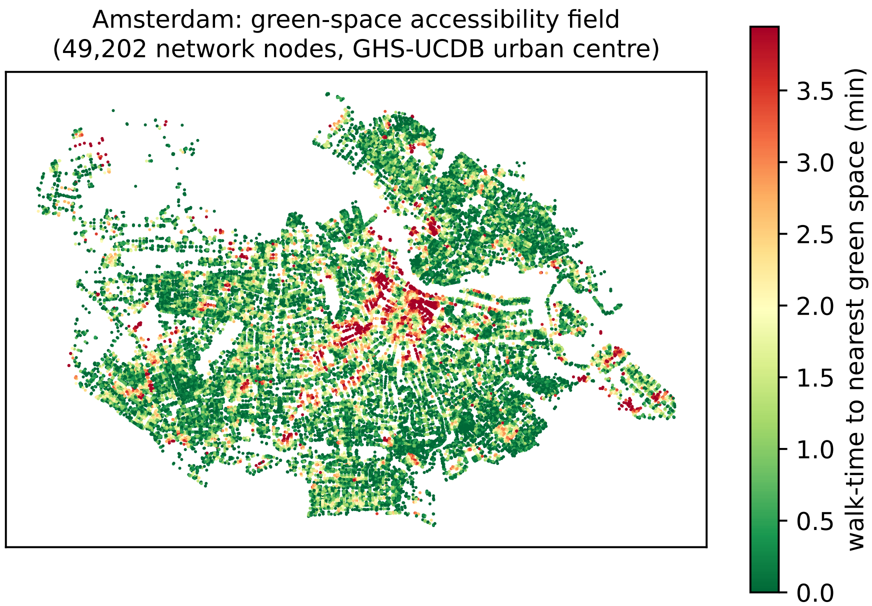

Figure 1: Green-space accessibility field for Amsterdam: each of the 49,202 pedestrian-network nodes inside the GHS-UCDB urban centre, coloured by network walking time to the nearest green space. Green marks well-served locations, red the access-poor pockets whose merging structure the resilience metric tracks.

and 5.0 km/h leaves the district AUC a near-exact rescaling of the 4.8 km/h values (Spearman $\rho = 0.98$ and 0.996; median AUC ratios 1.24 and 0.96, matching the expected 4.8/4.0 and 4.8/5.0). The field is defined over *network nodes*, each weighted equally; it is a property of the pedestrian network's spatial reach to green space, not a population-weighted or distributional measure. Our metric therefore speaks to one facet of SDG 11.7, the network's capacity to keep locations close to green space and how that capacity degrades under disruption, and not to the target's population-coverage or inclusivity dimensions, which would require census and land-use data we do not use here.

Resilience metric. Let $\mathrm{Dgm}(f)$ be the baseline diagram. A disruption of intensity $\rho \in [0,1]$ removes a fraction ρ of network edges; we recompute the field and diagram on the disrupted network and define the degradation

$$D(\rho) = \mathop{\mathbb{E}}_{r} W_1\big(\mathrm{Dgm}(f),\, \mathrm{Dgm}(f_{\rho,r})\big),$$

averaged over r replicate disruptions. We consider three disruption families: uniformly random edge removal (percolation-style), targeted removal of high-betweenness edges, and hazard removal of edges in low-elevation zones. These are deliberately *stylised stress tests*, not simulations of specific disasters: random removal probes generic structural robustness (and, being spatially uniform, does *not* reproduce the spatially clustered damage of a real flood or earthquake), targeted removal probes the worst case of losing critical links, and the hazard proxy adds a first, mild topographic signal. We use them to compare places under a common, reproducible perturbation, and we are correspondingly cautious in reading the results as real-disaster resilience (Limitations). Exact edge betweenness is $O(V|E|)$ and intractable at city scale, so the targeted scenario ranks edges by betweenness approximated from $k = 500$ sampled pivot sources. The resilience of a place is summarised by the area under the curve $\mathrm{AUC} = \int_0^{\rho_{\max}} D(\rho)\, d\rho$ (normalised by the ρ-range) and the critical transition ρ^*, the smallest ρ at which $D(\rho)$ first reaches half its maximum. We use the half-maximum as a scale-free, monotone summary of *where* the degradation curve turns over: unlike a fixed absolute threshold it adapts to each place's own range, and unlike the inflection point it is robust to the curve's noise. It is a descriptive locator of the transition, not a claim of a sharp percolation threshold, and (like AUC) it depends on the chosen disruption range $[0, \rho_{\max}]$; we therefore compare ρ^* only within a fixed range.

Homology dimension and denoising. On real street networks the sublevel filtration's finite H_1 is nearly empty: a street graph is close to planar and contains few triangles, so the network's cycles are never filled and almost all H_1 classes are essential (infinite) and carry no finite persistence. In Amsterdam, for example, the baseline diagram has 6,449 finite H_0 classes but only a single finite H_1 class (Fig. 2). This is a joint property of our *representation* (a node-valued walk-time field on the near-planar pedestrian graph) and the *sublevel filtration*, not a universal fact about green-space access. A different construction, for instance a Vietoris–Rips filtration on straight-line distances or a 2-complex that fills faces, would populate H_1 and could shift the signal into loops. Under our construction the finite information lives in H_0 (the merging of well-served regions), so we compute $D(\rho)$ on H_0 and treat the H_1 scarcity as a modelling consequence to revisit with alternative filtrations, not as evidence that access "deserts" are absent. To keep the exact Wasserstein matching tractable on city-scale diagrams we discard H_0 classes with lifetime below 1 walk-minute. This cut is principled rather than arbitrary: a one-minute difference in walk time is below the resolution at which the constant-speed (4.8 km/h) field is meaningful, so sub-minute classes are measurement noise, not access structure. Its effect is small and was tested directly (Persistence-threshold sensitivity, below): varying the cut over 0.5–2.0 minutes preserves every sign and the significance of the three headline descriptors, changing only reported magnitudes.

Districts and inference. Statistical inference rests not on five cities but on intra-urban *districts*: we tile each urban-centre boundary with a uniform H3 hexagonal grid (resolution 8, ≈ 0.7 km^2 cells) and compute, per district, its local resilience summaries and its local morphology (intersection density, circuity, orientation entropy, mean segment length, and a green-space fragmentation index—the number of distinct green-space patches per square kilometre of green area within the district, computed on the green space clipped to each hex so that it varies between districts rather than being a per-city constant collinear with the fixed effects). We then fit

$$\mathrm{AUC}_i = \beta^\top \mathbf{m}_i + \alpha_{c(i)} + \varepsilon_i,$$

an ordinary-least-squares regression of district AUC on morphology $\mathbf{m}_i$ with *city fixed effects* $\alpha_{c(i)}$ absorbing all *additive between-city* differences (any confound constant within a city, such as climate, planning regime or data vintage). They do not remove within-city confounding or effects that vary across districts inside a city; we address those separately with explicit controls and spatial models below. The five cities serve as a typological overlay, not as the inferential sample. This design also sidesteps the cross-city comparability problem of AUC. Because $\alpha_{c(i)}$ absorbs each city's overall AUC level, which scales with network and diagram size, the morphology coefficients are identified purely from *within-city* variation, and the inference never rests on comparing absolute AUC across cities. Only the descriptive city typology, not the regression, would need a size normalisation.

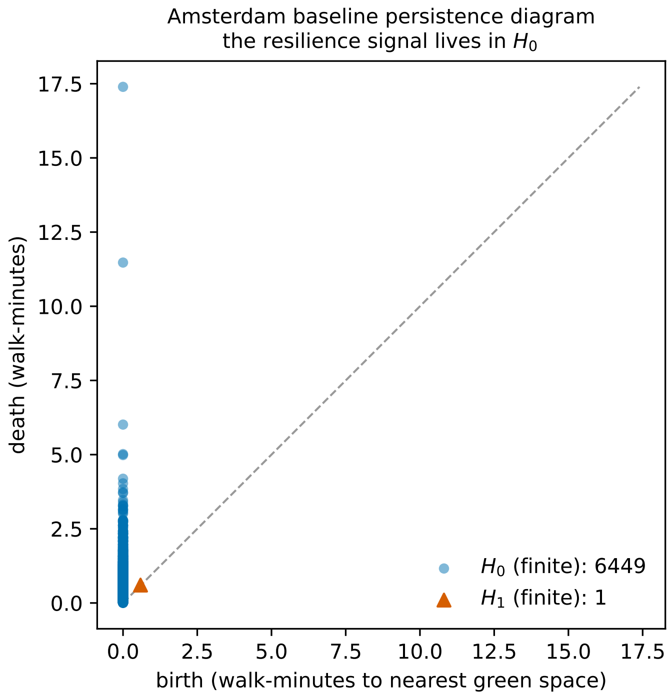

Figure 2: Persistence diagram of the Amsterdam baseline accessibility field. The finite part is carried almost entirely by H_0 (6,449 classes—the merging of well-served regions as the walk-time threshold rises) with only a single finite H_1 class, motivating an H_0-based resilience metric.

Table 1: District coverage (H3 resolution 8). Inference rests on $n = 2{,}603$.

City	Amsterdam	Barcelona	İstanbul	Bogotá	Phoenix	Total
Districts	227	147	729	538	962	**2,603**

5 Case studies

We study five cities chosen to span street morphology, planning regime and Global North/South divide: İstanbul (organic, hilly, polycentric, seismic), Barcelona (Cerdà grid with superblock interventions), Amsterdam (compact, canal, walkable), Bogotá (dense informal–formal mix), and Phoenix (low-density grid sprawl). Boundaries are the GHS-UCDB urban-centre polygons; homonymous centres (e.g. Barcelona, Spain vs. Venezuela) are disambiguated by largest true (equal-area) extent.

6 Results

The random-disruption analysis yields 2,603 qualifying districts (Table 1); only 2.6% have zero AUC. District mean AUC ranks Amsterdam as the most resilient and İstanbul/Bogotá as the least (Table 2); Phoenix has the earliest breakdown ($\rho^* = 0.16$). The district-mean ranking and the whole-city ranking below agree at the extremes (Amsterdam most resilient, Bogotá least) but not in the middle—Barcelona is mid-ranked by district mean yet most resilient at the whole-city scale—a reminder that AUC is comparable across cities only in broad ordering, not level.

The headline result is the district-level regression (Table 4), fit over the full disruption grid

Table 2: City typology (descriptive). Higher mean AUC $\Rightarrow$ access topology changes more under disruption $\Rightarrow$ *less* resilient.

City	mean AUC	mean ρ^*	mean H_0 total persistence
Amsterdam	4.46	0.25	13.11
Phoenix	6.68	0.16	8.08
Barcelona	7.67	0.28	13.63
İstanbul	7.75	0.20	12.84
Bogotá	8.02	0.24	16.90

(8 intensities $\times 10$ replicates). With city fixed effects and $n = 2{,}603$, three of five morphology descriptors are significant, two of them at $p < 10^{-15}$. Because the coefficient sign is on AUC (higher AUC = less resilient), a negative coefficient means the descriptor is associated with *greater* resilience. More circuitous networks (coef -10.16) and longer street segments (-0.048) are associated with more resilient green-space access, whereas higher orientation entropy ($+1.01$) is associated with less (Fig. 3). Intersection density is negatively signed, consistent with local redundancy aiding resilience, but is not significant ($p = 0.22$), so we treat it as at most suggestive. Green-space fragmentation is likewise not significant once it varies within cities and city fixed effects are included—plausibly because access in our field is set by the *nearest* green-space point, which is largely insensitive to how the remaining patches are split, and because the inclusive patch definition (Section 4) mixes very different green-space types into one count. Districts nest within five cities and are spatially contiguous, so their residuals are not independent; Appendix A treats this in full (city-clustered standard errors, a Moran's I diagnostic, and estimated spatial-error and spatial-lag models that Anselin LM tests and AIC strongly prefer to OLS). The two strongest effects, circuity and mean street length, survive every one of these treatments; orientation entropy survives the spatial models but not city clustering at five cities, which we weigh below.

External validity: replication across eight cities. Five cities give only five clusters, so the cluster-robust test above is underpowered. We therefore extended the analysis to three further cities on different continents—Singapore (planned, tropical), Nairobi (rapidly growing, mixed-formality) and Vienna (compact, central European)—for a 3,999-district, eight-city sample, and refit at a common reduced disruption grid (Table 3). The three headline signs replicate: more circuitous routing (-6.41) and longer segments (-0.046) predict greater resilience and higher orientation entropy ($+1.85$) less, all significant at $p < 10^{-11}$. Crucially, with eight clusters instead of five, *all three*—now including orientation entropy ($p = 0.025$)—survive city-clustered standard errors, indicating that its earlier failure was largely a small-cluster power problem rather than a fragile effect. Intersection density remains non-significant. (Two candidate cities, Copenhagen and Melbourne, were dropped because the OpenStreetMap place query did not resolve their full urban extent.)

The whole-city scale gives an *illustrative* typology, not a second inferential result: with only five cities it carries no statistical weight, and we report it to visualise the district-level signal, not to rank cities definitively. Computed on the same UCDB-clipped networks, the degradation curves $D(\rho)$ rise smoothly and order the cities consistently with the district means (Fig. 4). Because city-level AUC is computed on the full-city diagram (thousands of H_0 classes) it is two to three orders of magnitude larger than the per-district means of Table 2 and scales with network size; it is *not* normalised, so only the relative ordering—never the level—should be read, and a size-normalised cross-city AUC is left to future work. On that ordering, city-level AUC is lowest, and so most resilient, for Barcelona (620) and Amsterdam (898), and highest for İstanbul (2440) and Bogotá

Table 3: External-validity replication on eight cities (n = 3,999 districts; original five plus Singapore, Nairobi, Vienna) at a common reduced disruption grid, with city fixed effects. The three headline signs replicate, and with eight clusters all three—including orientation entropy—survive city-clustered standard errors. Sig.: $^{***}p < 0.001$, $^{*}p < 0.05$.

Morphology descriptor	coef	p (classical)	p (8-cluster)
circuity	−6.41	3×10^{-12}	1.6×10^{-4} ***
mean street length	−0.046	9×10^{-23}	8×10^{-18} ***
orientation entropy	+1.85	3×10^{-25}	0.025 *
intersection density	−0.023	0.70	0.88

Table 4: District fixed-effects regression, AUC $\sim$ morphology + C(city), n = 2,603, random-disruption scenario at the full disruption grid (8×10). Classical OLS standard errors; see Table 6 for city-clustered inference. Sig.: $^{***}p < 0.001$.

Morphology descriptor	coef	std. err.	p-value
circuity	−10.156	1.167	5.5×10^{-18} ***
mean street length	−0.0476	0.0058	3.7×10^{-16} ***
orientation entropy	+1.015	0.200	4.4×10^{-7} ***
intersection density	−0.091	0.074	0.22
green-space fragmentation	-2.0×10^{-6}	4.2×10^{-6}	0.64

(2828), with Phoenix in between (2053). The cities we would informally call compact, Barcelona and Amsterdam, retain more of their access topology under disruption than the large, sprawling or polycentric ones; we describe this pattern with the measured morphology descriptors rather than with "walkability," which we do not model directly. Under extreme removal ($\rho = 0.5$) the İstanbul and Phoenix curves turn down as the network fragments so heavily that unreachable nodes drop out of the finite diagram, a signature of near-total collapse rather than of recovered access.

Robustness (summary). The morphology relationship is stable across every alternative specification we tried; Appendix A gives the full results and Fig. 5 summarises them. Briefly: the three headline signs and their significance ($p < 10^{-3}$ or better) survive a *targeted* betweenness disruption, the coarse 5×3 and full 8×10 disruption grids, H3 resolutions 7/8/9 (a 28× range of district counts), persistence thresholds from 0.5 to 2.0 walk-minutes, a halving and doubling of the betweenness pivot count, and the inclusion of four district-level controls (green-space area, road hierarchy, relief and city-centre distance). Intersection density is the one descriptor that moves: significant under some specifications but not the full-grid headline, so we treat it as the weakest, grid-sensitive effect throughout. The one disruption family that does *not* carry the signal is the spatially clustered real-flood hazard, which we treat as a distinct question (below).

Green-space definition sensitivity. Because our inclusive green-space set mixes reliably public parks with possibly-private gardens, fenced grass and impenetrable woods, we re-ran the analysis on a strict, accessibility-conservative subset: only `leisure=park`, `recreation_ground` and `place=square`, dropping wood, grass and gardens. This is a large cut (in Amsterdam, from 23,657 features to 369), yet the district AUC is largely preserved (Spearman 0.82 against the inclusive set on the same reduced grid), and the regression is unchanged in substance: circuity (-8.08, $p = 4 \times 10^{-9}$),

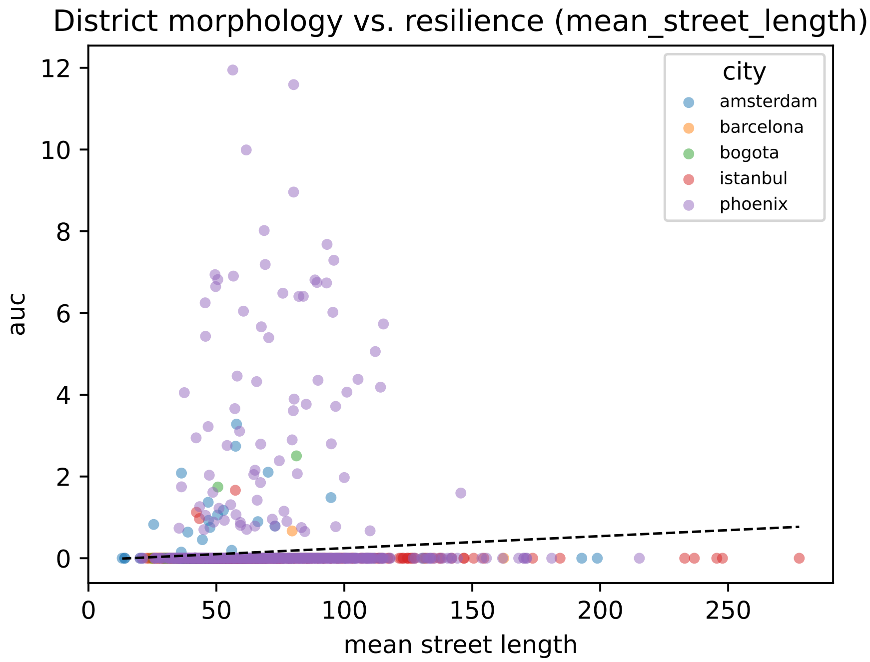

Figure 3: Headline morphology↔resilience relationship. Each point is one of the 2,603 districts, coloured by city; the dashed line is the pooled least-squares fit. Higher AUC indicates a district whose green-space access topology degrades more under disruption (less resilient).

mean street length (-0.048, $p = 2 \times 10^{-12}$) and orientation entropy ($+1.21$, $p = 2 \times 10^{-7}$) keep their sign and significance, and intersection density remains the weak, non-robust term. The morphology result is thus not an artefact of counting inaccessible green as accessible.

Criterion validity: benchmark against an accessibility measure. If the topological machinery is to be worth its cost it should track a simpler measure of access loss. We benchmark it, per district, against a *distance-based* accessibility-degradation measure, namely the increase in mean walk-time to green space under the same disruption (a proximity-accessibility degradation of the kind used throughout the field). As a second reference we use a *connectivity* degradation, the increase in the fraction of nodes left unable to reach any green space; this is the natural non-persistent counterpart, since H_0 of a sublevel filtration is single-linkage connectivity. The level of comparison is what settles the question. Our inference identifies morphology from *within-city* variation (city fixed effects), so the matching validity question is also within-city, once each city's overall AUC level (which scales with network size) is removed. Within city, the topological AUC correlates positively and highly significantly with the distance benchmark ($\rho = +0.36$, $p < 10^{-79}$, positive in all five cities; Table 5), so the metric tracks a simple accessibility-degradation ground truth at the level at which it is used. The naive pooled correlation is much weaker ($\rho = +0.07$) only because cross-city AUC scale swamps it, a Simpson's paradox driven by the same scale the fixed effects absorb. The benchmark also holds when access is *population-weighted* rather than node-uniform: weighting each node by the open GHS-POP residential-population grid [15] (JRC, 1 km, 2020; the same GHSL family and Mollweide grid as our urban-centre boundaries) gives an almost identical within-city correlation ($\rho = +0.35$, $p < 10^{-76}$). So the topological signal is not an artefact of weighting locations equally, and it speaks, at least for graded access, to the population-coverage

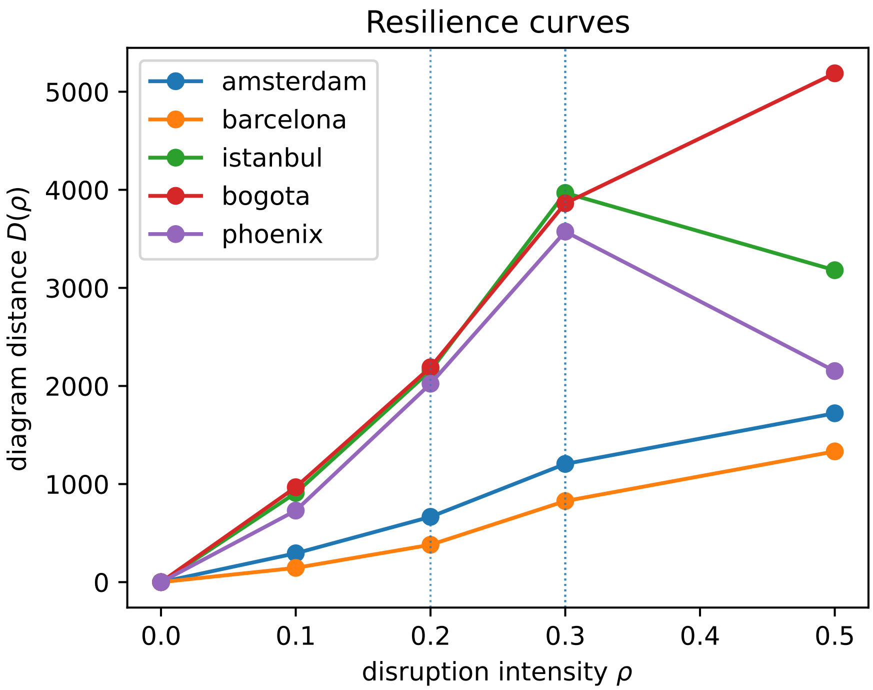

Figure 4: City-level resilience curves $D(\rho)$ for all five cities (random-disruption scenario), computed on the UCDB-clipped networks. Dotted vertical lines mark each city's critical transition ρ^*.

dimension of SDG 11.7 and not only to bare network reach. Two qualifications remain. The metric adds value rather than echoing the benchmark: controlling for both baselines and city fixed effects, the three headline morphology descriptors stay significant ($p < 10^{-6}$), so it carries configurational information the simple measures miss. And the convergent validity is with *graded* access loss, not with *disconnection*. Within city, AUC correlates near zero-to-negative with connectivity degradation, because disconnected nodes leave the finite diagram. The metric is thus validly read as the resilience of the *connected* served structure, and should be paired with a disconnection count for total access loss; we show in the Limitations that neither a naive finite cap nor extended persistence fixes this cleanly (both recover the connectivity sign only by sacrificing the graded signal), so we keep the two channels separate and scope the claim accordingly.

7 Discussion

The results give an interpretable, architecture-facing reading of resilience. The robust effects are circuity and segment length: districts with more circuitous routing and longer segments retain more of their well-served structure when edges are removed, because alternative paths to green space survive the loss of any one link. Denser intersections point the same way (locally redundant networks should be more resilient) but the effect is not statistically robust here, so we do not lean on it. Districts with more disordered street orientation (higher entropy) are more brittle. That the signal is carried by H_0 rather than H_1 has a concrete planning meaning. In green-dense cities the structure that matters is not isolated "deserts" surrounded by served land but the connectivity of the served regions themselves, and that connectivity is what disruption erodes.

For design, the reading is that *path redundancy*, not raw proximity, is what protects access to

Table 5: Criterion validity: Spearman correlation between the topological district AUC and the distance-based accessibility-degradation benchmark, per city and pooled within city (ranks taken within each city). Positive in every city; the within-city pooled value is what matters given the city fixed effects, and the naive pooled value is depressed by cross-city AUC scale.

Scope	n	Spearman ρ (AUC, distance degradation)
Amsterdam	227	+0.47 ***
Bogotá	538	+0.16 ***
İstanbul	729	+0.10 *
Barcelona	147	+0.05
Phoenix	962	+0.03
Within-city pooled	2,603	**+0.36** ***
Naive pooled	2,603	+0.07

nature under stress: the descriptors associated with resilience (circuitous routing, longer segments) are hallmarks of networks offering several comparable ways to reach the same green space, so that losing one link rarely isolates a served region. This reframes an SDG 11.7 design brief from "put a park within x metres" toward "ensure the walk to it survives the loss of individual streets"—a morphological, architecture-scale lever (block structure, connectivity, detour availability) rather than a purely land-use one. It also cautions against reading a tidy grid as automatically resilient: in our sample it is the less ordered, more redundant fabrics that degrade most gracefully.

Limitations. We group the caveats by the kind of claim they qualify.

Inference and spatial dependence. The design is cross-sectional and observational, so the regression identifies *associations* between morphology and topological resilience, not causal effects. Districts are also not independent: they nest within five cities and are spatially contiguous. City fixed effects absorb additive between-city differences, but two residual problems remain. First, the 2,603 districts furnish only five clusters for cross-city inference; clustering the standard errors by city (the level at which disruption is applied) widens the intervals substantially (Table 6). Under this conservative treatment the two strongest effects—circuity ($p = 6 \times 10^{-4}$) and mean street length ($p = 8 \times 10^{-8}$)—remain significant, while orientation entropy, significant at the district level, does not ($p = 0.26$). This last failure is largely a small-cluster power problem: in the eight-city extension (Table 3), with eight clusters, orientation entropy *does* survive city clustering ($p = 0.025$). With only five clusters, cluster-robust inference is underpowered, so the classical p-values in Table 4 understate uncertainty and are best read as within-sample descriptive strength; the eight-city replication is the more reliable cross-city test. Second, the residuals retain weak but statistically significant positive spatial autocorrelation within *every* city (global Moran's I on the first-ring H3 adjacency runs from 0.06 in Phoenix to 0.18 in Barcelona, all with permutation $p \leq 0.03$; Table 7). We addressed this directly by estimating spatial-error and spatial-lag models on a within-city weights matrix (Table 8): both absorb strong spatial dependence ($\hat{\lambda} = 0.55$, $\hat{\rho} = 0.53$), yet the three headline descriptors keep their sign and remain significant at $|z| \geq 5$, and intersection density in fact turns significant. The morphology signal is therefore not an artefact of spatial autocorrelation. On external validity, the eight-city extension (Table 3) shows the three signs replicating on three further cities across different continents, with all three surviving the now better-powered eight-cluster test; broader replication (more cities, and cities with sparser OpenStreetMap coverage than our set) remains the natural next step.

Omitted variables. Morphology is only one of many district attributes that could be associated with resilience. We control for the four that our data support—green-space area, road hierarchy, relief and city-centre distance (Table 9)— and the headline morphology effects survive their inclusion, so the association is not explained away by these four (which is weaker than ruling them out as causes). But population and built density, land-use mix, park-level size and quality, socioeconomic status and pedestrian-infrastructure quality would require external data (e.g. census and GHSL built-up layers) and remain uncontrolled; city fixed effects remove their between-city component but not their within-city variation, so some residual omitted-variable bias may persist.

Shared origin of predictor and outcome. Both the morphology descriptors and the resilience outcome are computed from the *same* street network, so part of the association is structural rather than a clean independent-variable/dependent- variable relationship: a district cannot have, say, low circuity and a routing topology that behaves as if it were highly circuitous. We therefore read the regression as characterising *which* formal properties co-vary with resilient access, not as an external cause acting on an independent outcome. Two observations temper the concern: the morphology descriptors explain only a modest share of the variance ($R^2 \approx 0.21$ with controls), so the outcome is far from a deterministic re-encoding of the predictors; and the accessibility field also depends on the green-space configuration, which is exogenous to street form. Still, a design that predicted one city's resilience from another's morphology would be needed to break the shared-origin coupling entirely.

What the metric measures, and how far it is validated. $D(\rho)$ is a *proposed proxy* for access resilience. We offer evidence of two kinds. Its *construct validity* is supported by its behaviour—it rises monotonically with disruption, is stable to field perturbations [5], and yields a morphology relationship that survives disruption model, district resolution, denoising threshold, pivot count, spatial-error/lag modelling and four omitted-variable controls. On *criterion* validity, the benchmark against a distance-based accessibility-degradation measure (Table 5) is positive at the level at which the metric is used. Within city, the unit the fixed effects identify from, the topological AUC correlates $+0.36$ ($p < 10^{-79}$) with the benchmark and is positive in all five cities, while remaining non-redundant (the morphology signal survives controlling for it). That validity has a definite scope: it holds for *graded* access degradation, not for total *disconnection*. Within city, AUC correlates near zero-to-negative with connectivity degradation, because unreachable nodes leave the finite diagram, so the metric measures the resilience of the *connected* served structure and should be paired with a disconnection count for total access loss. We tried to fold disconnection into a single scalar, and the result is instructive. A finite cap ($\tau = 60$ min) recovers the connectivity sign ($-0.10 \rightarrow +0.02$ on a two-city check) but erodes the graded-distance correlation ($+0.41 \rightarrow +0.06$), and a principled extended-persistence diagram, which gives the disconnected components finite coordinates in a separate channel, behaves the same way on Amsterdam (connectivity $+0.31$ but distance only $+0.05$), because the high-persistence component classes dominate a single Wasserstein match. Graded resilience and disconnection are, then, two channels that a single scalar conflates; reporting them separately, as we now do, is the right treatment. A joint multi-channel metric with a full head-to-head against an established measure (a population-weighted two-step floating catchment degradation, say, or observed post-disruption service loss) is the natural next step. Two further caveats: AUC depends on network and district size and on the number of H_0 classes, so it is comparable across cities only in relative ranking, not in absolute level (city fixed effects absorb this scale in the regression), and ρ^* depends on the chosen disruption range. The down-turn of the İstanbul and Phoenix curves at $\rho = 0.5$ reflects finite nodes dropping out of the diagram under near-total fragmentation, not recovered access, and marks the upper edge of the metric's useful range. Green spaces are treated as equivalent point sources, ignoring size, quality and public/private status.

Disruption and data. Random removal is spatially uniform and does not reproduce the clustered failures of a real flood or earthquake; targeted removal addresses criticality but not spatial correlation. The real 100-year flood scenario we add (above) makes this concrete: it materially affects only Vienna and does not carry the distributed-disruption morphology signal, showing that resilience to a localised hazard is a genuinely different quantity—one this paper does not claim to explain. Earthquake and multi-hazard disruption models are left to future work. Targeted betweenness is approximated from k=500 sampled pivots, as exact $O(V|E|)$ betweenness is intractable at city scale; the ranking is stable in the pivot count: on the Amsterdam network, halving or doubling k (to 250 or 1,000) leaves the edge ranking essentially unchanged (Spearman $\rho = 0.985$ against k=500, with 77%/90% overlap of the top-decile edges)—so the targeted scenario is not an artefact of the pivot count. Results use a persistence-thresholded H_0 metric, so reported magnitudes are denoised estimates; varying the threshold over a 4× range (0.5–2.0 walk-minutes) preserves every coefficient sign and keeps the three headline descriptors significant at $p < 10^{-6}$, and the full disruption grid confirms the signs, significances, ranking and ρ^*. OSM completeness varies by city and is not quantified here, and the network-distance Vietoris–Rips variant of the filtration is left to future work.

Finally, "resilience" here is a *structural* property—the robustness of the access topology to synthetic edge removal, measured on a single cross-sectional snapshot per city—not a temporal recovery process; we do not observe how access rebuilds over time after a real disruption, and the term should be read in that narrower, topological sense.

Read together, these caveats narrow the claim: within these five cities, and under synthetic edge-removal disruption, street-network morphology is a consistent descriptive correlate of green-space access resilience—robust in sign across disruption models, district resolutions, spatial specifications and the added controls—rather than a causally identified or cross-city-generalisable driver.

8 Conclusion

We defined a topological metric for the resilience of green-space access, the Wasserstein degradation curve of persistent-homology diagrams under network disruption, and showed across 2,603 districts in five contrasting cities that street-network morphology is associated with how robustly a city's pedestrian network keeps its *connected* served locations close to nature. Two effects are robust and survive spatial-error and spatial-lag models: more circuitous routing and longer street segments are associated with greater resilience. A third, orientation entropy, is weaker, significant at the district level but not once errors are clustered by the five cities. The association is consistent rather than proven causal, the study is cross-sectional, and it rests on five cities (see Limitations). We are also precise about what the metric captures: the resilience of the *connected* access structure, validated within city against a distance-based accessibility benchmark, rather than total accessibility including outright disconnection. The mathematics of persistence turns a static SDG 11.7 indicator into a dynamic, form-aware one. Future work will extend the real-hazard analysis beyond river floods to earthquakes and multi-hazard scenarios, and add a network-metric Vietoris–Rips filtration, toward a resilience-aware accessibility standard for sustainable cities.

Reproducibility. All data are open (OpenStreetMap, GHS-UCDB) and the pipeline is open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI/`persim`, H3, `statsmodels`); every reported number is recorded in the project's results log and traces to a committed artifact.

Table 6: Sensitivity of the regression inference to how standard errors treat the nesting of districts within cities. Coefficients are unchanged; only the standard errors and p-values differ. HC3 is heteroskedasticity-robust; the last two columns cluster by city (five clusters). Circuity and mean street length stay significant under city clustering; orientation entropy does not.

Morphology descriptor	coef	SE (HC3)	SE (cluster)	p (cluster)
circuity	-10.156	1.397	2.946	5.7×10^{-4} ***
mean street length	-0.0476	0.0066	0.0089	7.7×10^{-8} ***
orientation entropy	$+1.015$	0.220	0.894	0.26
intersection density	-0.091	0.092	0.214	0.67
green-space fragmentation†	-2.0×10^{-6}	5.9×10^{-6}	6.1×10^{-7}	$< 10^{-2}$

†The apparent clustered significance of green-space fragmentation is a five-cluster artefact of the cluster-robust variance around a coefficient that is effectively zero ($|\text{coef}| < 10^{-5}$); it is not a substantive effect.

Table 7: Global Moran's I of the fixed-effects regression residuals within each city, using first-ring H3 adjacency as the spatial weights (999 permutations). Residuals show weak but significant positive spatial autocorrelation in every city, motivating the estimated spatial models in Table 8.

City	districts	Moran's I	p (perm.)
Amsterdam	226	0.080	0.031
Barcelona	147	0.178	0.001
Bogotá	538	0.142	0.001
İstanbul	729	0.112	0.001
Phoenix	962	0.058	0.003

A Robustness and sensitivity analyses

This appendix collects the diagnostics summarised in the Results. Throughout, the three headline descriptors are circuity, mean street length and orientation entropy.

Spatial dependence. Districts nest within cities and are spatially contiguous, so the fixed-effects OLS residuals are not independent. Global Moran's I on first-ring H3 adjacency is positive and significant in every city (Table 7), and Anselin Lagrange-multiplier tests on the OLS residuals decisively reject the non-spatial model (LM-error $p < 10^{-99}$, LM-lag $p < 10^{-125}$; the robust LM statistics favour the lag form, 146 vs. 26), with both spatial models cutting the AIC by more than 400 relative to OLS (14,351 $\rightarrow$ 13,925 for the error model, 13,877 for the lag model). We therefore estimate spatial-error and spatial-lag models (Table 8). Both absorb strong dependence ($\hat{\lambda} = 0.55$, $\hat{\rho} = 0.53$), yet the three headline descriptors keep their sign and stay strongly significant ($|z| \geq 5$). City-clustered standard errors (Table 6), the most conservative treatment of the five-cluster design, leave circuity and mean street length significant but not orientation entropy.

Disruption scenarios. Under *targeted* disruption (removal of high-betweenness edges, ranked by k=500-sampled betweenness) every coefficient sign in Table 4 is reproduced: circuity -5.77 ($p = 4.6 \times 10^{-10}$), mean street length -0.034 ($p = 2.7 \times 10^{-13}$), orientation entropy $+0.50$ ($p = 1.6 \times 10^{-3}$), and intersection density (not significant under the full random grid) reaching $p = 0.022$ (-0.135). Absolute AUC values are lower (Amsterdam mean AUC 2.03 vs. 4.46 under random) but the ranking is unchanged.

Table 8: Estimated spatial models (maximum likelihood, `spreg`) with city fixed effects, on a block-diagonal within-city first-ring H3 weights matrix ($n = 2{,}602$ after dropping one island). Coefficients on AUC with z-statistics. The three headline descriptors keep their sign and remain strongly significant under both the spatial-error and spatial-lag specifications, despite substantial estimated spatial dependence ($\hat{\lambda}, \hat{\rho} \approx 0.5$).

Morphology descriptor	OLS (FE) coef (z)	Spatial error coef (z)	Spatial lag coef (z)
circuity	−10.15 (−8.7)	−5.38 (−5.0)	−6.88 (−6.7)
mean street length	−0.048 (−8.2)	−0.033 (−5.8)	−0.034 (−6.5)
orientation entropy	+1.01 (+5.1)	+1.25 (+5.9)	+0.92 (+5.2)
intersection density	−0.09 (−1.2)	−0.21 (−2.9)	−0.16 (−2.5)
green-space fragmentation	not significant in any model		
spatial parameter	—	$\hat{\lambda} = 0.55$	$\hat{\rho} = 0.53$

Real flood-hazard scenario. We replace the earlier low-elevation proxy with a *real* 100-year river-flood scenario, using the open JRC Global Flood Hazard maps [7] (floodMapGL, water depth per ≈ 1 km cell): a node is flood-exposed if it lies in a cell with ≥ 0.5 m of inundation, and the hazard removal progressively cuts flood-exposed edges. Two honest findings follow. First, real 100-year exposure among these urban centres is strongly city-specific: it is material only in Vienna (23% of nodes, on the Danube floodplain) and negligible elsewhere (Phoenix 0.7%, Amsterdam 0.4%, Barcelona 0.3%; İstanbul, Bogotá, Singapore and Nairobi essentially 0%, their centres sitting off the modelled river-flood plain). Second, where the scenario bites (Vienna), the morphology signal established for *distributed* disruption does *not* cleanly carry over to this *spatially clustered* flood: in a pooled flood-scenario regression the morphology coefficients are non-significant, and within Vienna only circuity keeps a weak same-signed effect (Spearman −0.13, $p = 0.006$) while orientation entropy even reverses. This is consistent with, and concretely demonstrates, our caveat that resilience to network-wide disruption is a different quantity from resilience to a localised, geographically concentrated hazard; morphology governs the former, not necessarily the latter.

Disruption-grid and resolution sensitivity. A coarser 5×3 disruption grid leaves every sign unchanged and keeps the three headline descriptors significant at $p < 10^{-3}$ (circuity −8.99, mean street length −0.0478, orientation entropy +1.17), with the city AUC ranking and ρ^* values within ≈ 0.02 of the full 8×10 grid. The one descriptor that moves is intersection density: significant on the coarse grid (−0.199, $p = 8.6 \times 10^{-3}$) but not on the full grid ($p = 0.22$). Re-tiling at H3 resolutions 7 (485 districts) and 9 (13,782 districts) likewise leaves the three headline descriptors significant at $p < 10^{-3}$ with constant signs across a 28× range of district counts; magnitudes scale with cell size but no sign flips. Figure 5 summarises both axes.

Persistence-threshold sensitivity. Varying the H_0 denoising threshold over a 4× range leaves the headline unchanged: at 0.5, 1.0 and 2.0 walk-minutes the three descriptors keep their sign and stay significant at $p < 10^{-6}$ (circuity $-11.20/-10.16/-8.03$; mean street length $-0.050/-0.048/-0.042$; orientation entropy $+1.10/+1.01/+0.93$). Magnitudes shrink smoothly as more denoising is applied but never change sign, and intersection density stays weak (ns at 0.5 and 1.0, $p = 0.04$ at 2.0). The betweenness pivot count is equally inconsequential: on the Amsterdam network, halving or doubling k (to 250 or 1,000) leaves the edge ranking essentially unchanged (Spearman $\rho = 0.985$

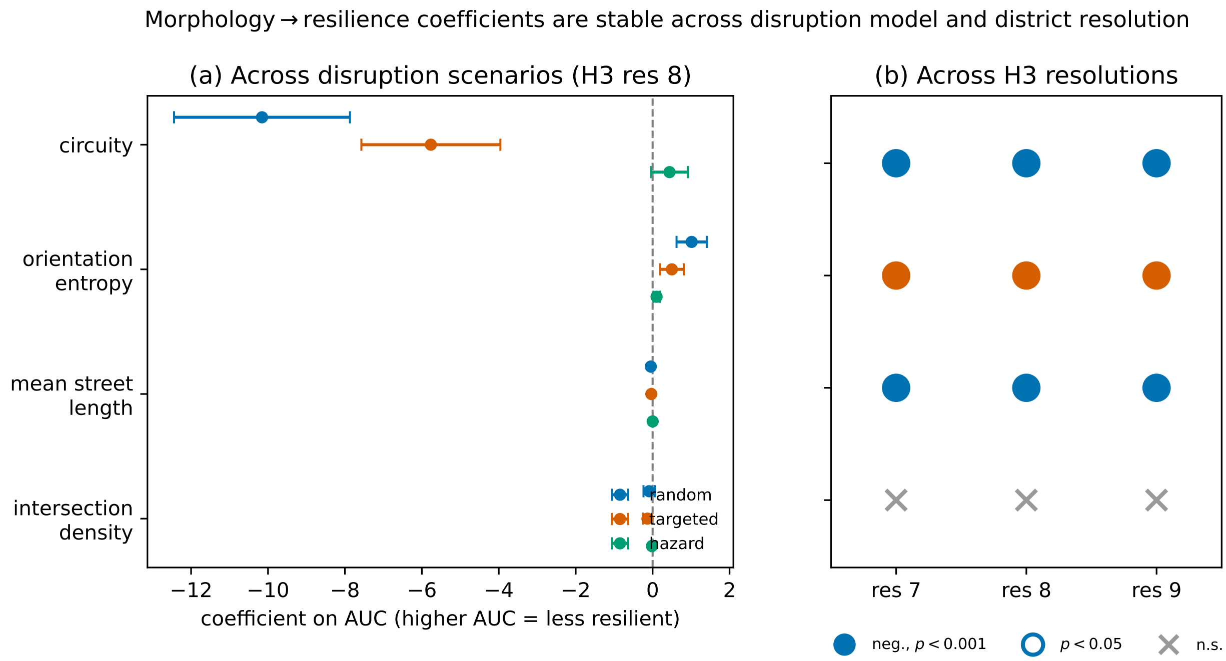

Figure 5: Robustness of the morphology↔resilience relationship. **(a)** Regression coefficients (with 95% confidence intervals) for each morphology descriptor under the three disruption scenarios at H3 resolution 8; the coefficient signs agree across scenarios, and the three headline descriptors (circuity, mean street length, orientation entropy) stay significant, while intersection density crosses zero under random disruption. **(b)** Sign and significance of each coefficient across H3 resolutions 7/8/9: the three headline descriptors stay significant with a constant sign, while intersection density, the weakest and grid-sensitive effect, is not significant at the headline full-grid specification.

against k=500, with 77%/90% overlap of the top-decile edges).

Omitted-variable controls. City fixed effects absorb only between-city confounds, so we add the district-level controls our data support—total green-space area, the share of edges on the trunk/primary/secondary/tertiary road hierarchy, within-district relief (elevation standard deviation), and distance to the urban-centre centroid—and refit (Table 9). Three of the four controls are themselves significant (more green area and greater centre distance with greater resilience, more major-road edges with less), yet the three headline descriptors keep their sign and stay significant at $p < 10^{-11}$ (circuity -9.55, mean street length -0.039, orientation entropy $+1.11$). Population and built density, land-use mix, socioeconomic status and pedestrian-infrastructure quality require external data and remain uncontrolled (Limitations).

Declarations

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Competing interests. The author declares no competing interests.

Table 9: Omitted-variable robustness: district regression before and after adding four controls, $n = 2{,}603$, with city fixed effects. The three headline morphology effects keep their sign and significance; the added controls explain additional variance (R^2 0.17 → 0.21). Sig.: $^{***}p < 0.001$, $^{**}p < 0.01$.

Variable	base coef (p)	+controls coef (p)
circuity	−10.16 ***	−9.55 ***
mean street length	−0.048 ***	−0.039 ***
orientation entropy	+1.01 ***	+1.11 ***
intersection density	−0.09	−0.10
green-space area (km^2)	—	−2.25 **
major-road share	—	+4.21 ***
relief (m)	—	−0.008
centre distance (km)	—	−0.107 ***

Data availability. All input data are open and publicly available. Pedestrian networks and green spaces were obtained from OpenStreetMap via OSMnx; urban-centre boundaries from the GHS Urban Centre Database (GHS-UCDB R2019A, European Commission Joint Research Centre); residential population from the GHS-POP grid (R2023A, 1 km) for the population-weighted validation; 100-year river-flood depth from the open JRC Global Flood Hazard maps (floodMapGL) for the real-flood scenario; and elevation data from the public AWS Terrain Tiles service. No new data were collected. The derived artifacts underlying every reported figure and table (per-district resilience and morphology tables, regression outputs, and city-level resilience curves) are recorded in the project's results log, released with the code at `https://github.com/navigaai/nnj-sdg11-topology`.

Code availability. The complete analysis pipeline is implemented in open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI, `persim`, H3, `statsmodels`) and is publicly available at `https://github.com/navigaai/nnj-sdg11-topology`, together with the configuration files and the results log that reproduce every reported number.

Author contributions. S. Emekci (ORCID 0000-0002-5470-6485) conceived the study, developed the method and software, performed the analysis, and wrote the manuscript.

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